🛡️ Module 07: Perimeter Defence Evasion Techniques

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# Exercise 02: Identifying and Bypassing a Firewall
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Scenario

Host-based and network firewalls inspect incoming and outgoing packets against configured rule lists, dropping unauthorized traffic and blocking diagnostic probes like ICMP traceroute requests. However, misconfigured firewalls often permit specific service ports (e.g., HTTP port 80, FTP port 21) or fail to block advanced TCP SYN probe techniques.

Penetration testers leverage specialized techniques such as **Firewalking** and raw packet probing with **Hping3** to analyze packet decay, discover open transit ports, and map active listening services behind filtering boundaries. By executing Nmap script engine (NSE) firewalk probes (**sudo nmap --script=firewalk --traceroute 172.19.19.7**) and TCP SYN incrementing probes (**sudo hping3 -S 172.19.19.7 -c 100 -p ++1**), security analysts can determine the exact filter boundaries and identify accessible ports for targeted exploitation.

In this exercise, we enable Windows Firewall on the Web Server machine (**172.19.19.7**), observe traceroute blocking on Parrot Security, perform Nmap firewalking to uncover **8** open ports, and probe port responses across filtered ranges using Hping3.

Objective

The objective of this lab is to help students learn how to:
- Configure host-based Windows Firewall settings across all network profiles on the Web Server
- Test target reachability using standard ICMP ping (**ping 172.19.19.7**) and analyze UDP/ICMP traceroute filtering (**traceroute 172.19.19.7**)
- Execute advanced Nmap firewalking scripts to determine gateway control rules and identify open ports (**sudo nmap --script=firewalk --traceroute 172.19.19.7**)
- Identify the total count of open ports (**8**) reported during active Nmap firewalk discovery
- Perform incremental TCP SYN packet probing using Hping3 (**sudo hping3 -S 172.19.19.7 -c 100 -p ++1**) to distinguish open vs. filtered ports

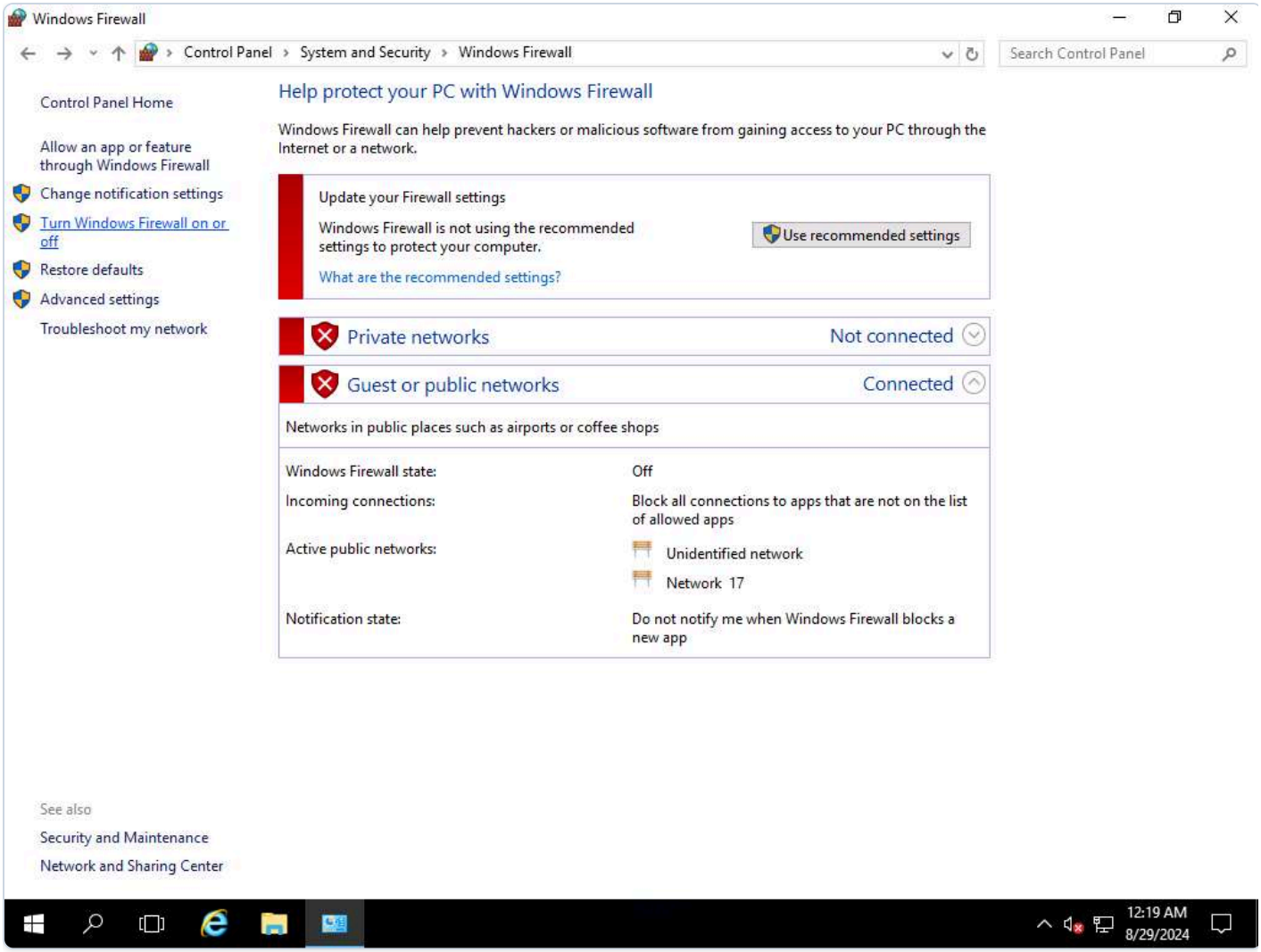
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Task 1: Enable Windows Firewall and Perform Firewalking Reconnaissance

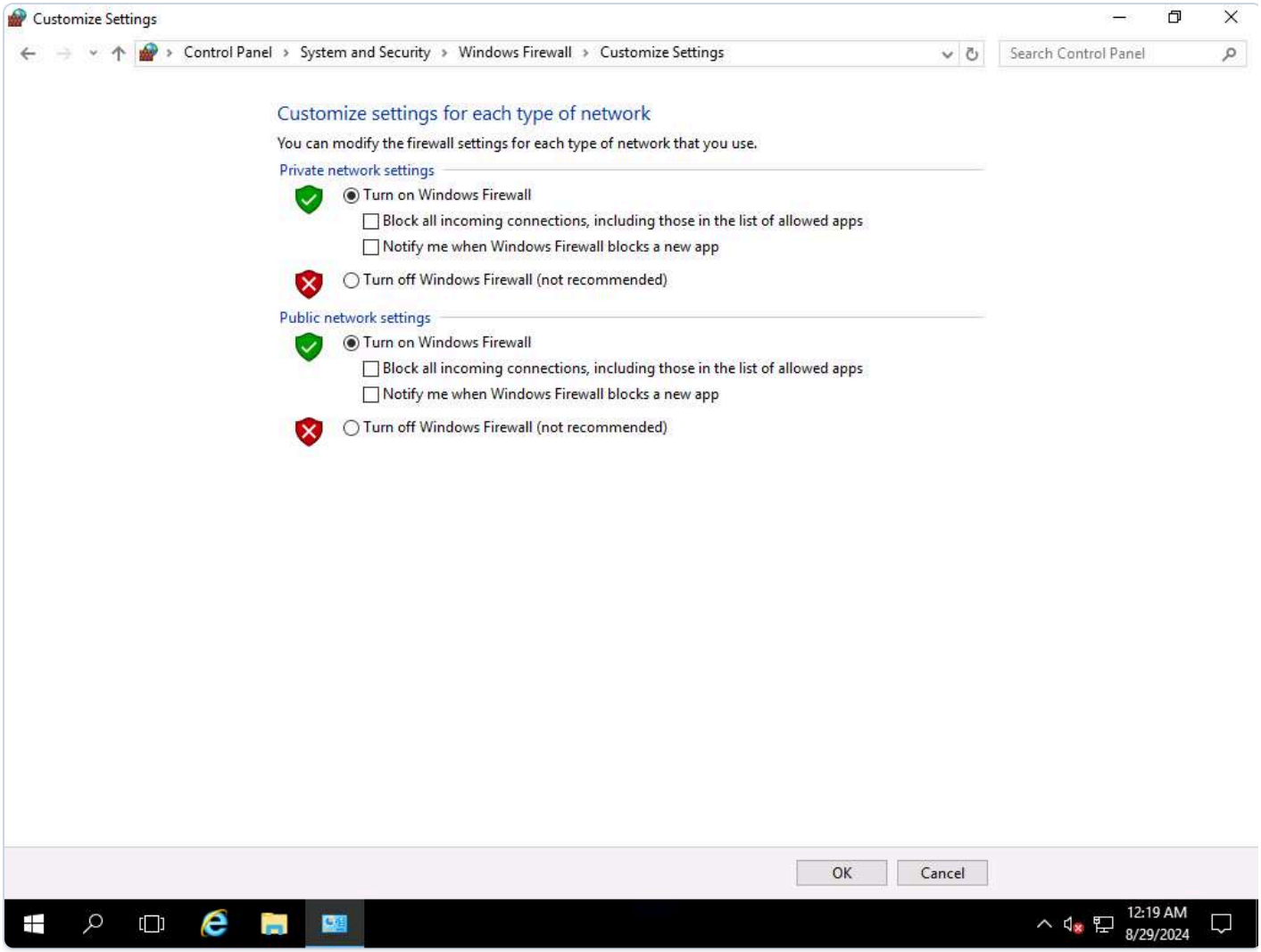
In this task, we turn on Windows Firewall on the Web Server target, initiate connectivity and traceroute tests from Parrot Security, run Nmap firewalking to discover open host ports, and execute custom TCP SYN probes using Hping3.

Lab Tasks

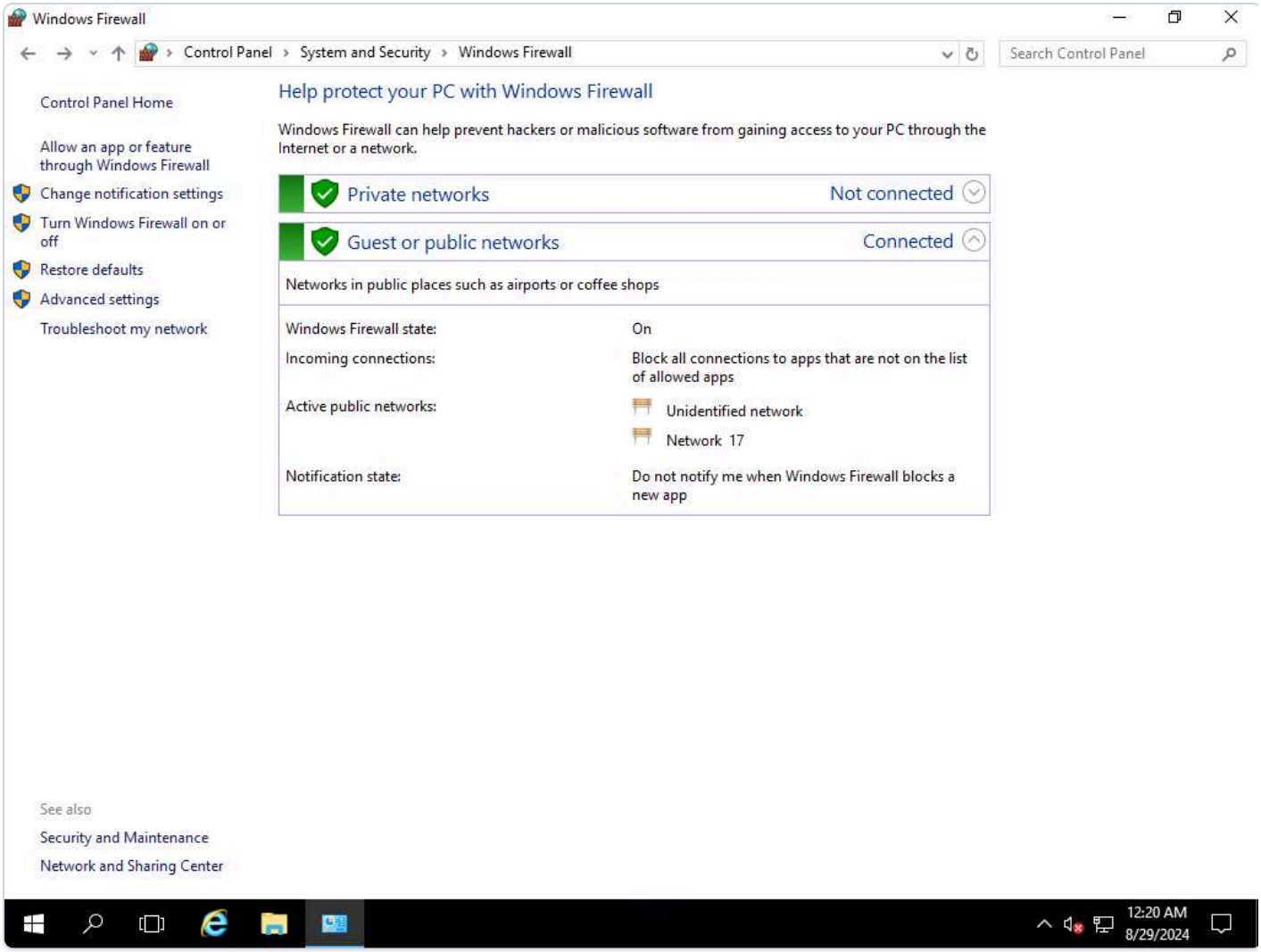
1. Click **Web Server** and press **Ctrl+Alt+Delete**. Log in with password **Pa$$w0rd**.
2. Open **Control Panel** $
   ightarrow$ **System and Security** $
   ightarrow$ **Windows Firewall**. In the left pane, click **Turn Windows Firewall on or off**.



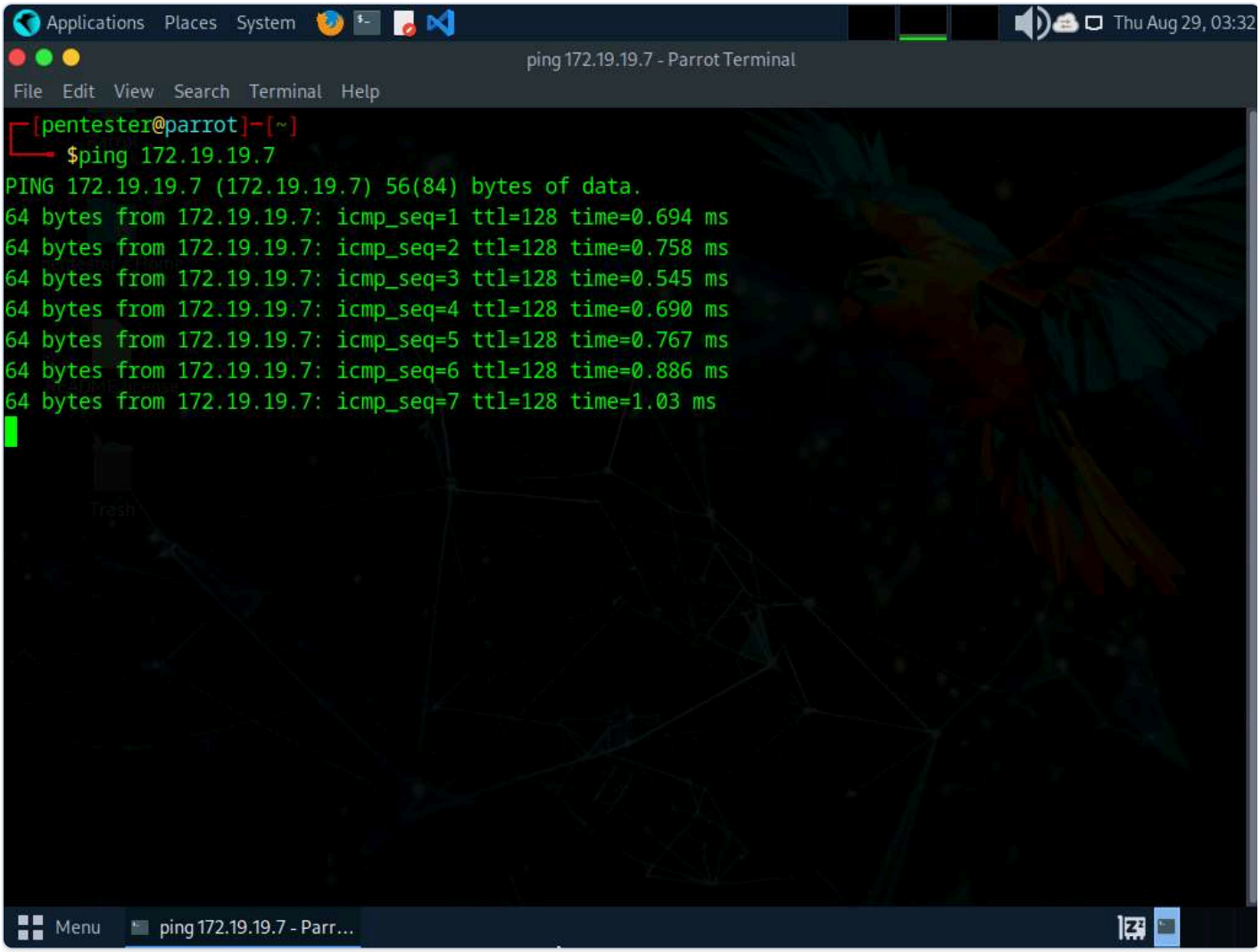
1. Select **Turn on Windows Firewall** under both **Domain/Private** and **Public** network location settings, then click **OK**.



1. Confirm that Windows Firewall is actively enabled on the Web Server.



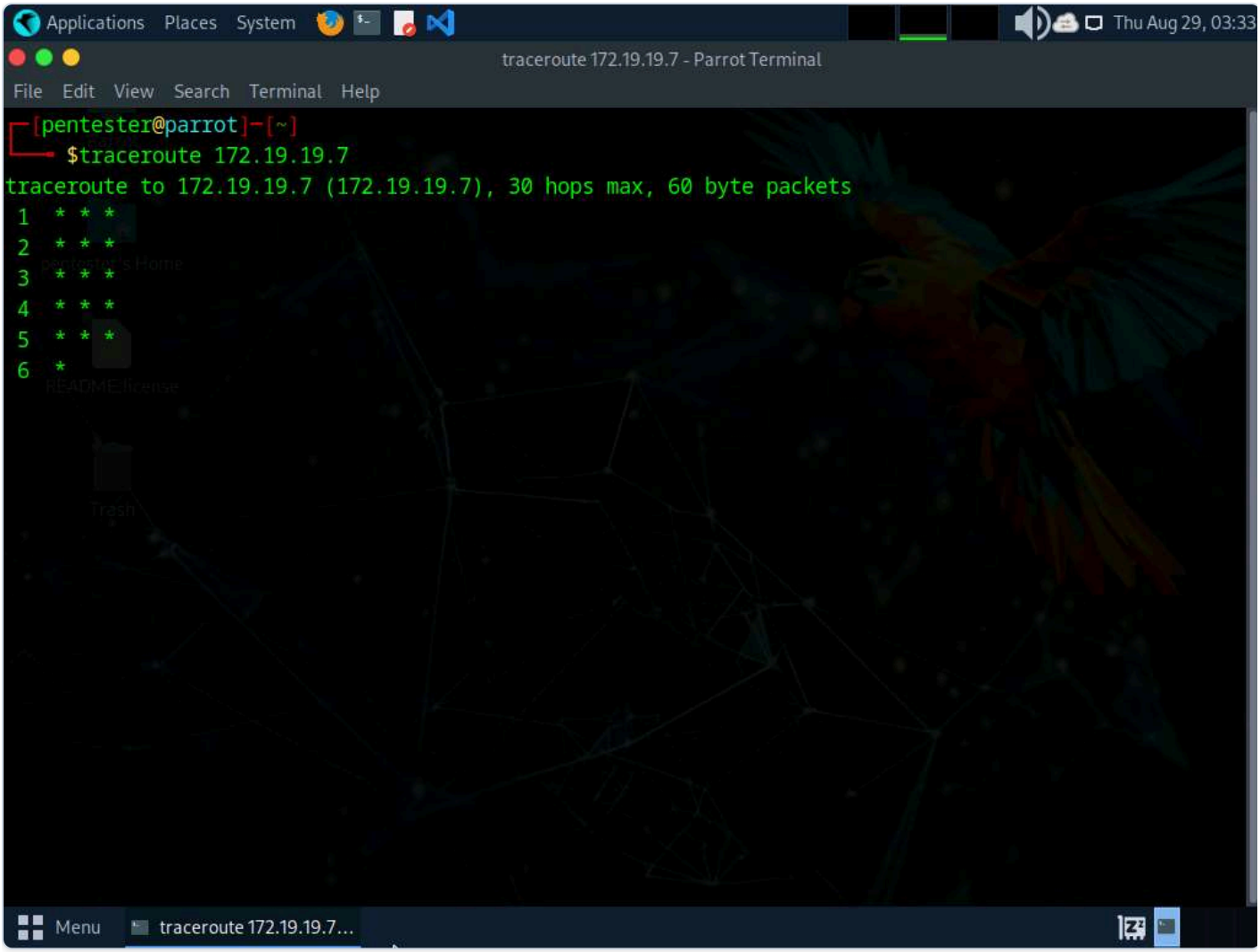
1. Switch to **Parrot Security** (**toor**). Open a terminal and test ICMP connectivity to the target:
   ping 172.19.19.7
   (Press Ctrl+C after verifying replies).



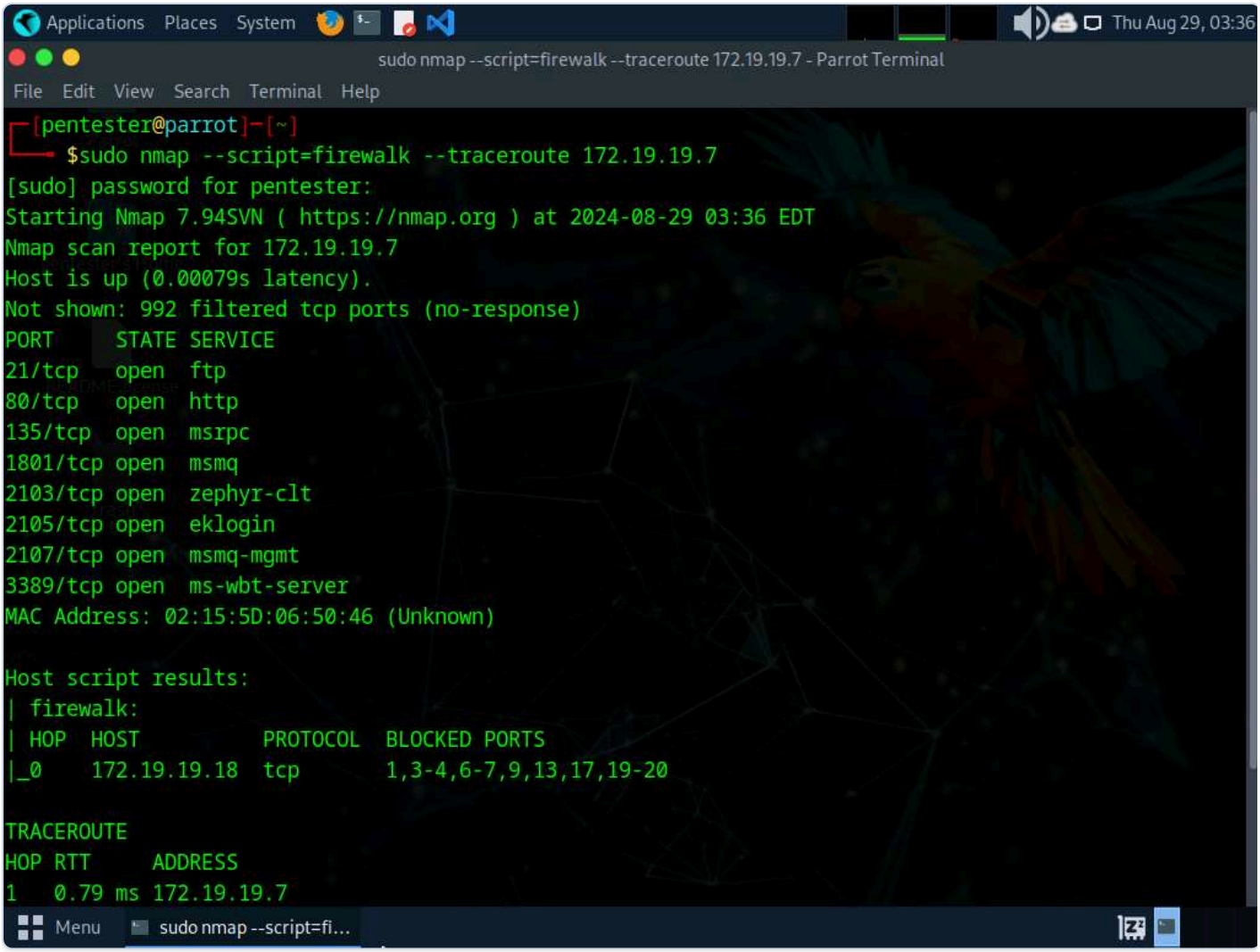
1. Execute a standard traceroute scan against the Web Server:

   traceroute 172.19.19.7

   Observe that packet path requests time out (`* * *`) because Windows Firewall blocks standard UDP/ICMP traceroute probes. Press Ctrl+C to abort.



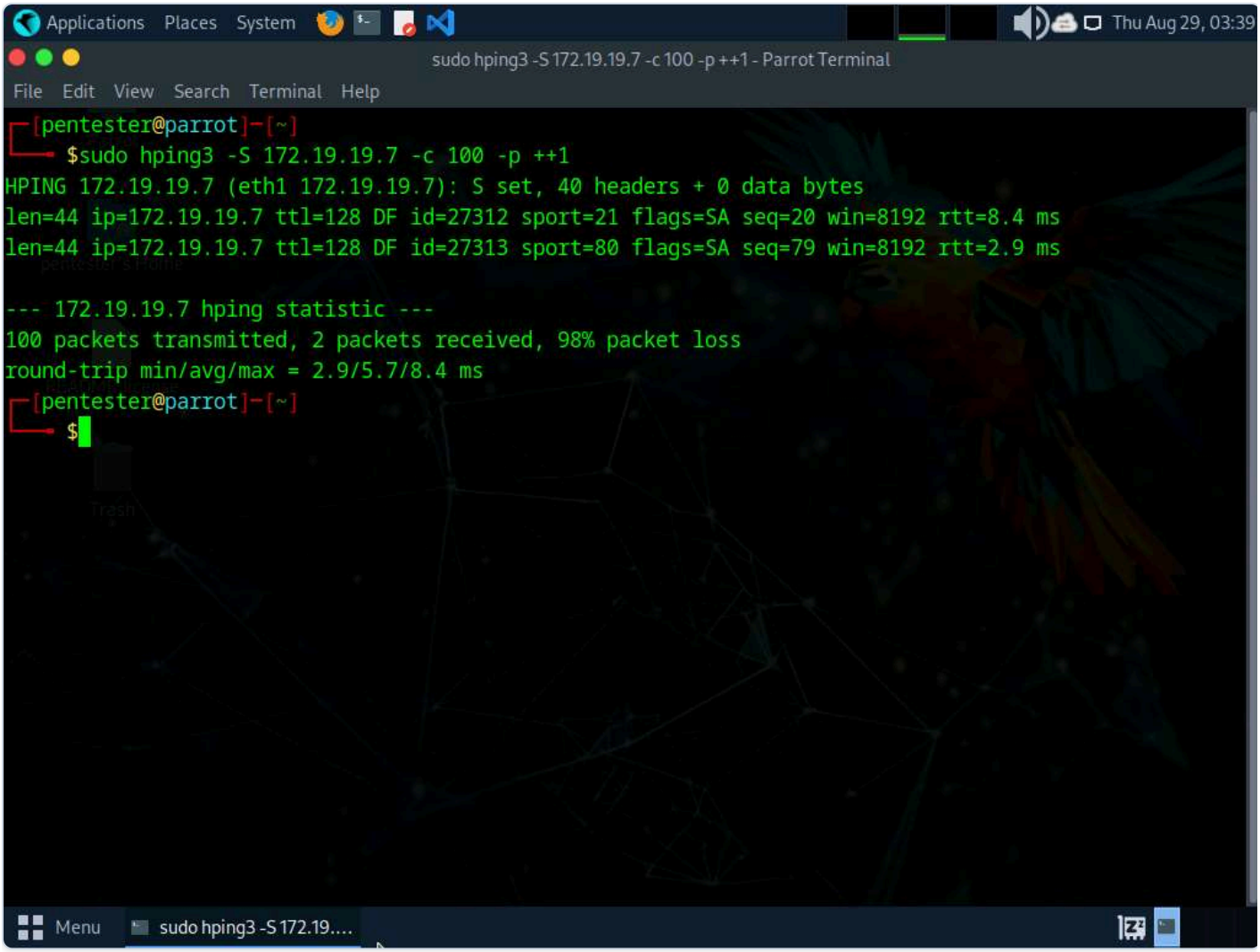
1. Launch an advanced Nmap firewalking traceroute scan:
   sudo nmap --script=firewalk --traceroute 172.19.19.7
   *Analyze the output to identify host script results, traceroute hops, and the total number of open ports on the victim server (**8** open ports).*



1. Execute incremental TCP SYN probes across the first 100 ports using Hping3:

   sudo hping3 -S 172.19.19.7 -c 100 -p ++1

   Observe that out of 100 transmitted packets, 98 packets are filtered by the firewall, while packets sent to open ports (e.g., TCP 21, TCP 80) successfully receive SYN-ACK responses.



1. Close all active terminal and configuration windows.

This concludes the lab exercise.

IMPORTANT: ❓ **Question 7.2.1.1** In Web Server machine, turn on the firewall and run ping, traceroute, nmap and hping3 commands to bypass firewall from Parrot Security machine on Web Server (172.19.19.7). Enter the number of open ports that are open while performing nmap scan. [!SUCCESS] ✅ **Answer:** **8**